Design of a Rooftop Solar Power Plant for a Specific Site

A. Objective

Design and evaluate the technical and financial feasibility of a rooftop photovoltaic (PV) solar power plant for a real site. The project involves site assessment, solar panel selection, performance estimation, and profitability analysis.

1. Site Assessment

Evaluate the suitability of the selected site (e.g., M4 Building Terrace) for a PV installation.

- Determine the latitude, longitude, and altitude of the site.
- Obtain local solar irradiance data (GHI, DNI) from PVGIS, NASA POWER, or MGMY Enerji Atlası.
- Consider temperature, cloud cover, and annual sunlight hours.
- Area available: 100 m^2; consider tilt, azimuth, and shading.

2. Solar Panel and System Design

Select appropriate solar panels and system configuration.

- Choose PV module (e.g., 400 W, 1.9 m^2, 20% efficiency).
- Determine number of panels for 85% area utilization.
- Compute total system capacity and select inverter (DC/AC ratio 1.1–1.2).

3. Energy Output Estimation

Estimate annual generation using site irradiance and system efficiency.

Formula: $E_{annual} = P_{sys} \times PSH \times 365 \times \eta_{system}$

Where $\eta_{system} \approx 0.75–0.85$ (considering system losses).

4. Financial Analysis

Perform cost and profitability analysis.

- Installed cost: 700–1000 USD/kWp.
- Electricity price: 0.15 USD/kWh.
- Calculate annual savings, payback period, NPV, and IRR.

5. Environmental and Practical Considerations

Discuss operational issues: temperature derating, dust, maintenance, grid interconnection, and CO_2 reduction.

6. Deliverables and Grading

Submission must include calculations, plots, and a feasibility summary.

Section	Description	Weight (%)
Site Assessment	Data accuracy and solar resource evaluation	15
PV System Design	Panel/inverter selection and layout	25
Energy Calculation	Application of formulas and interpretation	25
Financial Analysis	Investment and payback accuracy	20
Realism & Presentation	Professional quality,	15

environmental insights